Abstract

As Large Language Models (LLMs) grow in size, quantization to 4-bit precision has become a standard method for efficient deployment. However, its impact on non-English languages is underexplored. We evaluate Llama-3-8B on English and Swedish benchmarks (NLI and WiC) across 16-bit, 8-bit, and 4-bit precisions. We find that 4-bit quantization causes a distinct "class collapse," specifically destroying the model's ability to detect neutral/ambiguous relationships in NLI tasks. To mitigate this, we implement Quantized Low-Rank Adaptation (QLoRA) by fine-tuning the 4-bit model on a translated Alpaca dataset. Our results show that QLoRA not only recovers general Swedish performance (increasing NLI F1 from 0.54 to 0.63) but specifically restores the decision boundaries for neutral classes. However, this comes at the cost of catastrophic forgetting in English lexical tasks.

1 Introduction

The widespread adoption of LLMs is hindered by their immense computational and memory requirements. Quantization, the process of reducing the precision of model weights (e.g., from 16-bit to 4-bit), offers a powerful solution, enabling deployment on consumer hardware (Dettmers et al., 2023). While effective for English, the impact of aggressive quantization on morphologically rich languages, such as Swedish, remains an open question. Furthermore, it is unclear if quantization errors are uniformly distributed or if they disproportionately affect specific linguistic capabilities, such as handling ambiguity.

In this work, we investigate the trade-off between efficiency and multilingual performance. Our contributions are threefold: (1) We evaluate Llama-3-8B on Natural Language Inference (NLI) and Word-in-Context (WiC) tasks using SuperLim, SuperGlue and Glue, utilizing constrained generation for robust evaluation. (2) We identify a "neutral class collapse" in 4-bit quantization, where the model loses the nuance required to classify non-binary relationships. (3) We demonstrate that fine-tuning with QLoRA (Dettmers et al., 2023) on a small Swedish instruction dataset successfully recovers these lost capabilities, albeit with trade-offs in English generalization.

2 Related Work

Quantization Post-Training Quantization (PTQ) techniques like LLM.int8() (Dettmers et al., 2022) and NF4 (NormalFloat 4-bit) (Dettmers et al., 2023) have allowed for massive reductions in memory usage. While Dettmers et al. (2023) demonstrated that 4-bit models preserve perplexity on English benchmarks, the degradation in downstream zero-shot tasks for less dominant languages is less documented.

Multilingual Adaptation Adapting English-centric models to other languages typically involves continued pre-training or instruction tuning. Cross-lingual transfer is often observed, but "tax" on the original language (catastrophic forgetting) remains a challenge when parameter budgets are tight (Artetxe et al., 2020).

3 Methodology

3.1 Models and Quantization

We use **Llama-3-8B-Instruct** (Dubey et al., 2024) as our base model. We evaluate three configurations using the `bitsandbytes` library:

- **16-bit (M1)**: The baseline loaded in BFloat16.
- **8-bit (M2)**: Quantized using LLM.int8().
- **4-bit (M3)**: Quantized using the 4-bit NormalFloat (NF4) data type with Double Quantization.

3.2 QLoRA Fine-tuning

To mitigate performance loss, we train a fourth configuration, 4-bit Adapter (M4). We freeze the M3 base model and attach Low-Rank Adapters (LoRA) to the query, key, value, and output projection layers (W_q, W_k, W_v, W_o). We use rank $r = 16$, $\alpha = 32$, and dropout 0.05.

We fine-tune on ALPACA-SWEDISH-REFINED (Neph1, 2023), a translated version of the Alpaca instruction dataset. We train on a subset of 5,000 samples for 1 epoch using the Paged AdamW optimizer to handle memory spikes during gradient checkpointing. The training dynamics (loss and accuracy) are visualized in Appendix Figure 1.

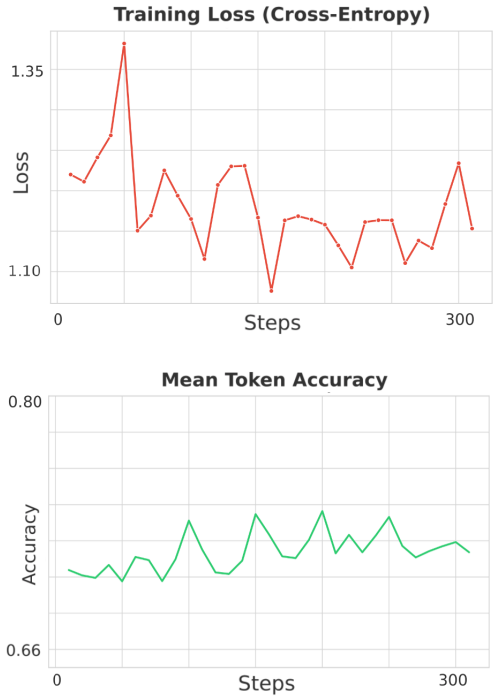

Figure 1: Training metrics for the M4 Adapter (Loss and Token Accuracy).

3.3 Constrained Evaluation Pipeline

We evaluate on four tasks (500 samples each): MNLI (English) from GLUE (Wang et al., 2018), Swenli (Swedish) from SuperLim (Berdicevskis et al., 2023), WiC (English) from SuperGLUE (Wang et al., 2019), and SweWic (Swedish) from SuperLim (Berdicevskis et al., 2023).

A common issue in evaluating quantized LLMs is their tendency to violate output formatting (e.g., outputting "The answer is Entailment" instead of "Entailment"). To ensure fair evaluation, we utilize the **Outlines** library (Willard and Louf, 2023). We construct a regex-guided generation pipeline that forces the model to select valid tokens corresponding to the task labels (e.g., `(Entailment|Neutral|Contradiction)`), eliminating parsing errors.

4 Experiments and Results

4.1 Quantization Degradation

Table 1 presents the aggregate results, which are further visualized in Figure 2. We observe that 4-bit quantization (M3) causes a significant drop in NLI performance. Swedish NLI drops from 0.62 (16-bit) to 0.54 (4-bit), a degradation of $\approx 12\%$.

The aggregate F1 scores mask a critical failure mode. As shown in the per-class breakdown (Table 2) and confusion matrices (Appendix Figures 4 and 5), the 4-bit model suffers from a collapse of the "Neutral" class. In English NLI, the F1 score for "Neutral" drops from 0.22 (16-bit) to **0.02** (4-bit). The model essentially becomes a binary classifier (Entailment vs. Contradiction), losing the ability to detect ambiguity. This suggests that the "nuance" bits are the first to be lost during quantization.

Model	NLI En	NLI Sv	WiC En	WiC Sv
16-bit (M1)	0.608	0.623	0.629	0.608
8-bit (M2)	0.590	0.602	0.631	0.583
4-bit (M3)	0.527	0.545	**0.646**	**0.642**
4-bit Adapter (M4)	0.545	**0.631**	0.531	0.632

Table 1: Macro F1 scores. Note the recovery in NLI Sv for M4, and the degradation in WiC En.

Task	Class	16-bit	4-bit	Adapter
NLI En	Entailment	0.79	0.79	0.70
NLI En	Neutral	**0.23**	**0.02**	**0.36**
NLI En	Contradiction	0.80	0.77	0.58
NLI Sv	Entailment	0.76	0.72	0.74
NLI Sv	Neutral	**0.32**	**0.12**	**0.39**
NLI Sv	Contradiction	0.80	0.79	0.76

Table 2: Per-class F1 scores showing the collapse of the Neutral class in 4-bit and recovery in Adapter.

Interestingly, the Word-in-Context (WiC) task appears robust to quantization. In fact, M3 (4-bit) slightly outperforms M1 (16-bit) on Swedish WiC (0.642 vs 0.608). Given the small sample size ($N = 500$), this may be attributed to statistical noise, or potentially a regularization effect where lower precision prevents overfitting to specific context artifacts.

4.2 QLoRA Recovery and Trade-offs

The QLoRA fine-tuned model (M4) successfully recovers Swedish capabilities. As visualized in Figure 3, Swedish NLI performance rebounds to 0.631, surpassing the 16-bit baseline. Crucially, M4 restores the "Neutral" class capability, achieving a Neutral F1 of 0.39 in Swedish (up from 0.12 in 4-bit), indicating that the adapter layers have learned to re-establish these decision boundaries.

While Swedish performance improves, we observe clear signs of catastrophic forgetting in English. While English NLI remains stable, English

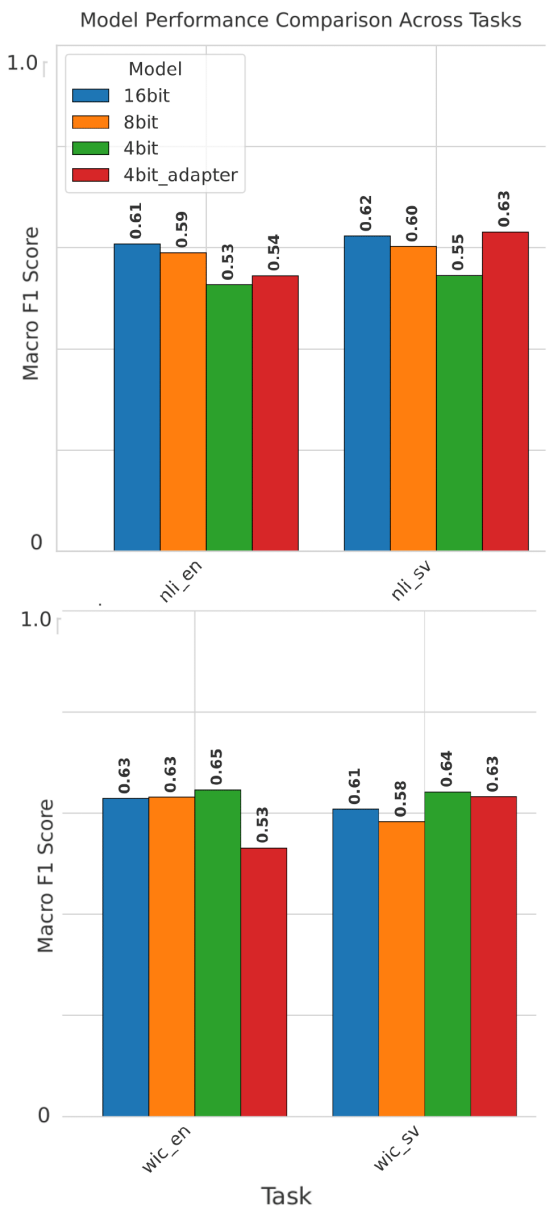

Figure 2: Performance comparison across all tasks. Note the drop in English NLI for 4-bit (M3) and the recovery in Swedish tasks for the Adapter (M4).

WiC performance drops significantly from 0.646 (M3) to 0.531 (M4). Since WiC requires precise lexical knowledge, this suggests the model has updated its internal representations to favor Swedish semantics at the expense of English lexical definitions.

5 Conclusion

Our study confirms that while 4-bit quantization makes LLMs efficient, it degrades performance

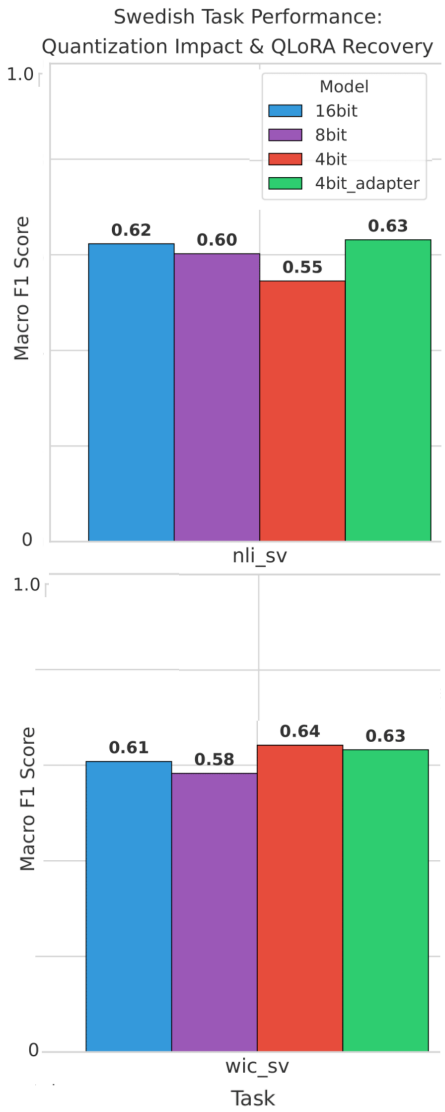

Figure 3: Impact of QLoRA fine-tuning (M4) on Swedish tasks, illustrating the successful recovery of performance lost to quantization.

non-uniformly, specifically effectively erasing the model's ability to handle neutral/ambiguous NLI cases. We demonstrated that QLoRA is a highly effective mitigation strategy, not only recovering general performance but restoring these complex decision boundaries. However, this adaptation incurs a "language tax", evident in the degradation of English lexical tasks. Future work should explore mixed-language instruction tuning to balance recovery with preservation.

Limitations

We used a translated dataset (Alpaca-Swedish) which may contain translation artifacts. The "Neutral" class baseline for Llama-3-Instruct was

already low (0.22) even in 16-bit, suggesting the base model itself struggles with the specific label distribution of the MNLI/SuperLim benchmarks, which quantization exacerbated.

References

Mikel Artetxe, Sebastian Ruder, and Dani Yogatama. 2020. On the cross-lingual transferability of monolingual representations. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 4623–4637. Association for Computational Linguistics.

Aleksandrs Berdicevskis, Gerlof Bouma, Felix Dürer, Ariel Ekgren, et al. 2023. Superlim: A swedish language understanding evaluation benchmark. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 8137–8153.

Tim Dettmers, Mike Lewis, Younes Belkada, and Luke Zettlemoyer. 2022. Llm.int8(): 8-bit matrix multiplication for transformers at scale. In *Advances in Neural Information Processing Systems*, volume 35, pages 30311–30324.

Tim Dettmers, Artidoro Pagnoni, Ari Holtzman, and Luke Zettlemoyer. 2023. Qlora: Efficient finetuning of quantized llms. In *Advances in Neural Information Processing Systems*, volume 36, pages 10088–10115.

Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Keshwam, Ahmad Al-dahle, et al. 2024. The llama 3 herd of models. *arXiv preprint arXiv:2407.21783*.

Neph1. 2023. Alpaca-lora-gpt4-swedish-refined. `https://huggingface.co/datasets/neph1/Alpaca-Lora-GPT4-Swedish-Refined`.

Alex Wang, Yada Pruksachatkun, Nikita Nangia, Amanpreet Singh, Julian Michael, Felix Hill, Omer Levy, and Samuel Bowman. 2019. Superglue: A stickier benchmark for general-purpose language understanding systems. In *Advances in Neural Information Processing Systems*, volume 32.

Alex Wang, Amanpreet Singh, Julian Michael, Felix Hill, Omer Levy, and Samuel Bowman. 2018. Glue: A multi-task benchmark and analysis platform for natural language understanding. In *Proceedings of the 2018 EMNLP Workshop BlackboxNLP: Analyzing and Interpreting Neural Networks for NLP*, pages 353–355.

Brandon T Willard and Rémi Louf. 2023. Efficient guided generation for large language models. *arXiv preprint arXiv:2307.09702*.

A Appendix

A.1 Confusion Matrices

Figures 4 and 5 visually confirm the class collapse. Note how the 4-bit model (third plot) predicts almost zero Neutral labels, while the Adapter (M4) partially restores the central column.

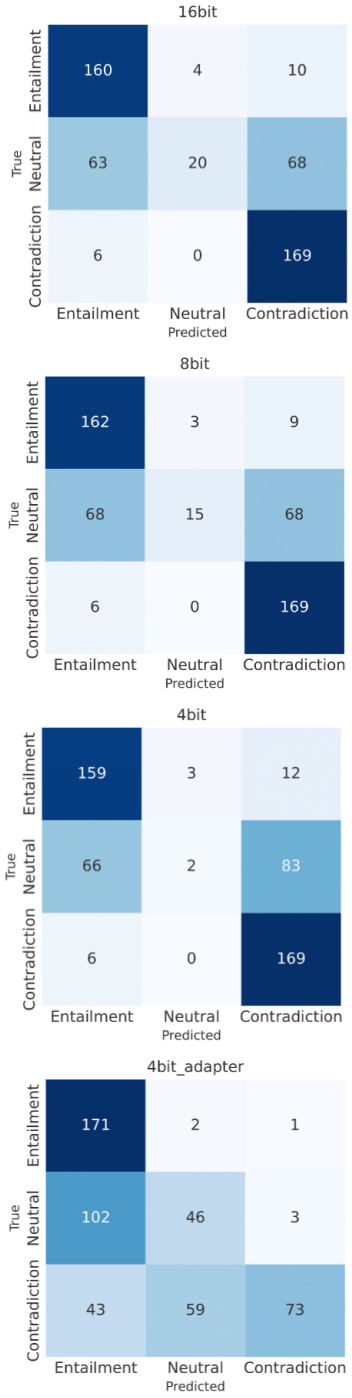

Figure 4: Confusion matrices for English NLI. 4-bit (M3) collapses the Neutral class.

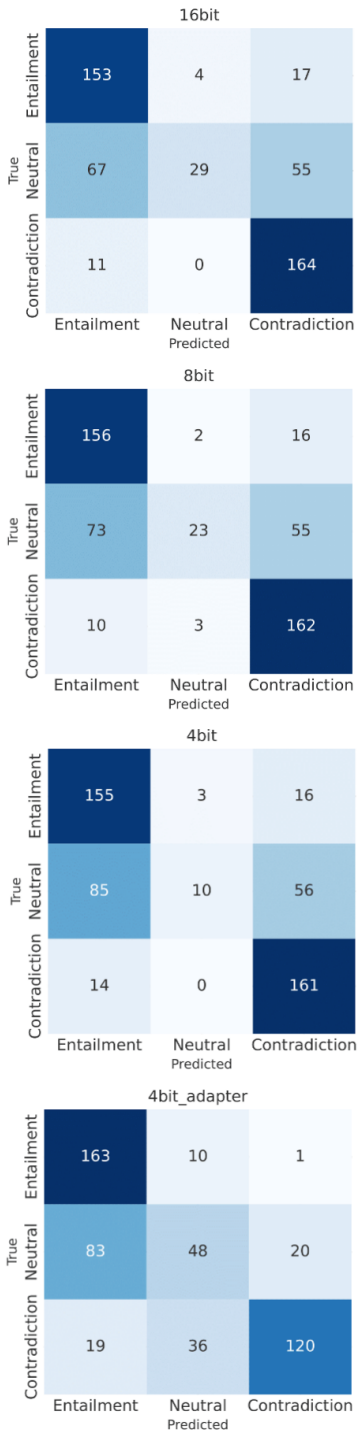

Figure 5: Confusion matrices for Swedish NLI showing similar recovery patterns with QLoRA.